

Tika Ram Gurung

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RESEARCH PROFILE

“Climate scientist with a background in cryosphere processes and regional climate modeling, investigating hydroclimate variability from a glacier to regional scales. My current work integrates numerical modeling and large datasets to understand human-climate interactions, with a focus on how soil moisture, irrigation, and moisture transport shape climate variability.”

EDUCATION:

Ph.D. Candidate in Climatology/Meteorology (Jan 2023 - present)

Expected Graduation: Summer 2026

Department of Earth and Atmospheric Sciences, University of Nebraska-Lincoln.

*Dissertation: **High-Resolution Climate Simulations and Impacts of Local Human Factors on Hydroclimate and Glacier Dynamics in High Mountain Asia (ongoing)***

M.S. in Glaciology at Kathmandu University, Nepal (2015)

Thesis: Reconstruction of annual mass balance of Rikha Samba Glacier, Nepal, from 1995 to 2013 using Energy-Mass Balance Model.

B.Sc. in Environmental Science (Hons) at Kathmandu University, Nepal (2013)

WORK EXPERIENCES:

Research Associate/Cryosphere Analyst at ICIMOD (2016-2020/2021-2022)

Led and supported Cryosphere research expeditions in high mountain glaciers (20) for glacier monitoring activities including installation/maintenance of automatic weather stations, temperature loggers, dGPS survey, played a key role in database and knowledge generation and dissemination, and contributed to the capacity building of regional member countries (Afghanistan, Bhutan, India, Pakistan) through a series of training related to field-based glacier monitoring and statistical analysis of the collected data.

PUBLICATIONS:

Gurung, T.R., Chen, L., Ali, S.H., Rowe, C., Arriola, F.M. and Ritzema, R.S., 2026. Evaluating the performance of high-resolution climate simulations over the Central Himalaya and Karakoram. *Climate Dynamics*, 64(3), p.120 (<https://doi.org/10.1007/s00382-025-08030-x>).

Sato, Y., Buri, P., Kneib, M., Miles, E.S., Sunako, S., **Gurung, T.R.**, Sakai, A., Pellicciotti, F. and Fujita, K., 2025. Modelling seasonal fluctuations and aspect characteristics of ice-cliff melt on the debris-covered Trakarding Glacier, Rolwaling Valley, Nepal Himalaya. *Journal of Glaciology*, 71, p.e33 (<https://doi.org/10.1017/jog.2025.17>).

Gurung, T.R. and Chen, L., 2024. Understanding the influence of soil moisture on heatwave characteristics in the contiguous United States. *Environmental Research Letters*, 19(6), p.064070 (<https://doi.org/10.1088/1748-9326/ad4dbb>).

- Baral, P., Allen, S., Steiner, J.F., **Gurung, T.R.** and McDowell, G., 2023. Climate change impacts and adaptation to permafrost change in High Mountain Asia: a comprehensive review. *Environmental Research Letters* (<https://doi.org/10.1088/1748-9326/acf1b4>).
- Gurung, T.R.**, Kayastha, R.B., Fujita, K., Joshi, S.P., Sinisalo, A. and Kirkham, J.D., 2023. A long-term mass-balance reconstruction (1974–2021) and a decadal in situ mass-balance record (2011–2021) of Rikha Samba Glacier, central Himalaya. *Journal of Glaciology*, pp.1-14 (<https://doi.org/10.1017/jog.2022.93>).
- Stumm, D., Joshi, S.P., **Gurung, T.R.** and Silwal, G., 2021. Mass balances of Yala and Rikha Samba Glacier, Nepal from 2000 to 2017. *Earth System Science Data*, 13(8), pp.3791-3818 (<https://doi.org/10.5194/essd-13-3791-2021>).
- Steiner, J.F., **Gurung, T.R.**, Joshi, S.P., Koch, I., Saloranta, T., Shea, J., Shrestha, A.B., Stigter, E. and Immerzeel, W.W., 2021. Multi-year observations of the high mountain water cycle in the Langtang catchment, Central Himalaya. *Hydrological Processes*, p.e14189 (<https://doi.org.libproxy.unl.edu/10.1002/hyp.14189>).
- Gilbert, A., Sinisalo, A., **Gurung, T.R.**, Fujita, K., Maharjan, S.B., Sherpa, T.C. and Fukuda, T., 2020. The influence of water percolation through crevasses on the thermal regime of a Himalayan mountain glacier. *The Cryosphere*, 14(4), pp.1273-1288 (<https://doi.org/10.5194/tc-14-1273-2020>).
- Gurung, T.R.**, Acharya, A., Dildar, M. and Joshi, S.P., 2020. Preliminary hydrometeorological data analysis of the Kabul River basin, Afghanistan: Need for a long-term glacier monitoring programme. *ICIMOD working paper* (<https://lib.icimod.org/record/34946>).

AWARDS:

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- Outstanding Graduate Award 2024:** Department of Earth and Atmospheric Sciences, University of Nebraska-Lincoln
- The Daugherty Water for Food Global Institute Graduate Research Award for 2024/2025,** University of Nebraska
- Cryosphere Monitoring Project (CMP) funded MS in Glaciology Scholar (2013-2015),** Kathmandu University

RECENT CONFERENCE ATTENDED:

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- Gurung, T.R.**, and Chen, L. (2026). From Lowlands to Mountains: Irrigation Impacts on South Asian Hydroclimate. Poster presentation at the 136th Annual Meeting of the Nebraska Academy of Sciences (NAS). Lincoln, Nebraska, April 24, 2026.
- Gurung, T.R.**, and Chen, L. (2026). Potential Impacts of Irrigation on Hydroclimatic Variability over High Mountain Asia. Poster presentation at the 106th Annual Meeting of the American Meteorological Society (AMS). Houston, Texas, January 25-29, 2026.
- Gurung, T.R.**, and Chen, L. (2024). High-resolution climate simulations of HMA. Poster presentation at the American Geophysical Union annual meeting (AGU), Washington D.C., December 9-13, 2024.
- Gurung, T.R.** and Chen, L. 2023: Understanding the Influence of Soil Moisture on Heatwave Characteristics in the United States, AGU Fall Meeting, San Francisco, December 11-15, 2023.
- Gurung, T.R.**, Steiner, J.S., Sherpa, T.C., and Joshi, S.P. 2023: Mass and Energy Balance of two Himalayan glaciers, Central Himalaya, NAS Annual Meeting, Lincoln, April 21, 2023.

RECENT TRAINING PARTICIPATED IN:

April 7-9, 2025	MPAS-Atmosphere Tutorial, NCAR (virtual).
July 17-21, 2023	WRF tutorial summer 2023, NCAR.
June 07-11, 2021	Spatial and temporal climate change analysis using CORDEX regional climate models over South Asia, ICIMOD.

SKILLS:

Coding & Analysis	R, Python, MATLAB, and GIS, GitHub, Linux-based environments, High-performance computing (HPC)
Modeling	Weather and Research Forecasting (WRF), Model for Prediction Across Scales (MPAS), Community Earth System Model (CESM), Energy Mass Balance Glacier Model, Noah-MP Land Surface Model
Machine Learning	Scikit-learn, statistical learning methods, time-series analysis, causal inference frameworks, neural network fundamentals
Data & Publishing	NetCDF, HDF5, ERA5/reanalysis products, LaTeX, MS Office
Field monitoring	Installation and maintenance of automatic weather stations, rain gauges, dGPS surveying, temperature loggers, surface radiation measurement systems
Language	Nepali (native), English (Fluent), Hindi (Excellent), Urdu (Good)

PUBLIC ENGAGEMENT

Summer 2025	Volunteer, UNL Earth and Atmospheric Sciences outreach events for elementary schools
Winter 2023 and 2024	Volunteer, Dinosaurs & Disasters, University of Nebraska State Museum

MEMBERSHIP IN PROFESSIONAL ORGANIZATIONS

American Geophysical Union
American Meteorological Society
Nebraska Academy of Sciences
European Geophysical Union

LEADERSHIP

President 2025/2026, Nepalese Students Association (NSA) at the University of Nebraska-Lincoln

PODCAST

University of Nebraska - Daugherty Water for Food Podcast on [The Cryosphere and Food & Water Security](#)

SERVICE TO PROFESSIONAL PUBLICATION

Journal Reviewer:
Environmental Monitoring and Assessment